Hamdan Ashfaq

Irvine & San Diego, CA · +1 (740) 915-5051 · hashfaq1@uci.edu · linkedin.com/in/hamdanashfaq

EDUCATION

University of California, Irvine & San Diego State University (Joint Doctoral Program) Aug 2026 – Present
Ph.D., Computational Science · Expected 2031 *Irvine & San Diego, CA*

- *Graduate coursework (in progress / planned):* Scientific Computing; Advanced Computational PDEs; Mathematical Modeling; Foundations of Deep Learning; Large Language Models; Quantum Physics; Modern Physics; Condensed Matter Physics.

Denison University Jan 2023 – May 2026
B.A. Computer Science & B.S. Data Analytics, Philosophy concentration *Granville, OH*

- **Cumulative GPA 3.51 / 4.0 · SAT 1510 / 1600**
- Departmental honor cords in both Computer Science and Data Analytics; Denison Phi Society Inductee, top 7% of Class of 2026.

TECHNICAL SKILLS

Programming: Python, C/C++, R, SQL; Git, Linux, Jupyter, RStudio.

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, TIMM; deep learning, self-supervised learning, CNNs, vision transformers, unsupervised clustering, dimensionality reduction.

Quantitative Methods: Time-series analysis, statistical and predictive modeling, mixed-integer linear programming, optimization (Gurobi), Monte Carlo and scenario analysis, numerical simulation.

Data & Computing: NumPy, Pandas, Matplotlib, OpenCV; HPC, CUDA, parallelized pipelines; PowerBI, Tableau.

RESEARCH EXPERIENCE

The Ohio State University, Center for Electron Microscopy & Analysis (CEMAS) May 2024 – Aug 2026
Visiting Fellow, Materials Science and Engineering (May 2026 – Aug 2026) *Columbus, OH*

- Developed multislice electron-scattering simulations (Kirkland TEMSIM) that generate physics-based nanodiffraction patterns from atomistic models of amorphous semiconductors, producing ground-truth data to train and validate machine-learning models in a regime where experimental labels do not exist.
- Built simulation-to-ML pipelines coupling forward electron-scattering physics with deep-learning classification of nanoscale structural motifs, addressing labeled-data scarcity in amorphous-material characterization.
- Ran large-scale simulation workflows on the Ohio Supercomputer Center, optimizing job scheduling and throughput across many-core clustered compute resources.

Machine Learning Research Intern (Aug 2025 – Dec 2025) *Columbus, OH*

- Developed time-series analysis frameworks for 4D-STEM data to track microstructural evolution in amorphous semiconductors across varying experimental conditions and thermal-processing parameters.
- Built predictive models identifying temporal patterns in structural transformations using sequential machine-learning techniques and autocorrelation analysis on high-dimensional experimental data.

Research Intern, Intel-funded OSU SURE-5 Program (May 2024 – May 2025) *Columbus, OH*

- Led computational analysis of 65,536 nanodiffraction patterns (multiple GB) using unsupervised machine-learning algorithms under full Intel Corporation sponsorship, achieving 80% classification efficiency on unlabeled 4D-STEM data.
- Developed supervised ML models to analyze atomic structures of amorphous semiconductors, optimizing 4D-STEM data processing and identifying prevalent structural patterns through unsupervised techniques.

Purdue University, SURF Research Program May 2025 – Aug 2025
Engineering Research Intern (advisor: Prof. Nikhilesh Chawla) *West Lafayette, IN*

- Engineered a deep CNN combining a ResNet-50 backbone with a U-Net++ inspired decoder to classify fishbone, lamellar, fine, and coarse SEM microstructures in Sn-58Bi solder alloy images, achieving 90.14% accuracy.
- Cut microstructure analysis time from hours of manual expert review to roughly 30 minutes of automated classification, accelerating characterization of solder joints used in electronic packaging.
- Developed SEM-specific preprocessing pipelines (ROI cropping, CLAHE contrast enhancement, intensity normalization, geometric augmentation) and raised image-processing throughput 30% via parallelized, optimized TensorFlow data pipelines.

SELECTED PROJECTS

Self-Supervised Classification of Electron Diffraction Patterns (Senior Capstone) Aug 2025 – May 2026

- Built a self-supervised learning pipeline (SimSiam with a ResNet-18 backbone) to learn representations from unlabeled 4D-STEM diffraction patterns of metallic glasses, detecting fold symmetry in amorphous structures without labeled training data; executed on the Ohio Supercomputer Center. **Awarded Best Senior Capstone Project, Department of Data Analytics.**

Power-Grid Unit Commitment and Dispatch Optimization Fall 2025

- Formulated and solved a mixed-integer linear program scheduling a thermal and renewable generation fleet over a 72-hour horizon on real PJM load and renewable-availability data, modeling startup costs, minimum stable output, renewable availability limits, and a 15% reserve-margin constraint across the fleet.
- Produced a base-case optimal objective of $27.96M and stress-tested the fleet across high-demand and low-renewable scenarios, identifying reserve shortfalls in 53 of 72 hours with a peak deficit near 233 MW as a quantified reliability signal.

Predictive Modeling of Organizational Outcomes on Panel Data Fall 2025

- Applied principal component analysis and k-means clustering to a 138-organization panel dataset spanning 1998 to 2015 to derive latent group types from twelve behavioral and network indicators.
- Trained a Random Forest classifier on the derived clusters plus organizational covariates to predict above- versus below-median annual outcome counts, reaching an AUC of 0.655 against a 0.5 random baseline, and benchmarked it against logistic regression.

Templated Route-Planning Engine in C++ Spring 2025

- Implemented Dijkstra shortest-path search over a templated adjacency-list graph with a hand-built binary-heap min-priority queue supporting O(log n) insert, extract-min, and decrease-key, giving O((n + m) log n) end-to-end runtime.
- Benchmarked the solver at 0.0128 seconds for a 99-vertex shortest path on large graphs, added fault-tolerant parsing that skips malformed records, and resolved an integer-overflow defect by templating identifiers to 64-bit types.

PUBLICATIONS & PRESENTATIONS

Journal Articles, Under Review

- Islam, M., Wang, M., Fantin, A., Yoo, G.H., Kim, J.Y., **Ashfaq, H.**, Park, E.S., Maass, R., Wang, Y., Fan, Y., Chung, H.S., Lee, S.C., Hwang, J. "Linking Topological Medium-Range Order and Density-Wave Coherence to Metallic Glass Ductility." *Acta Materialia* (under review, 2026). Contribution: Swin Transformer V2 classifier for weak rotational symmetries in amorphous nanodiffraction data.

Conference Presentations & Posters

- Patnaik, S., Ganju, E., Torbati Sarraf, H., **Ashfaq, H.**, Chawla, N. "Harnessing the Power of AI in 3D X-ray Microtomography." Poster, Purdue AI Research Showcase, West Lafayette, IN, 2026.
- **Ashfaq, H.**, Wu, J., Chawla, N. "Imaging and Machine Learning of Defects in Semiconductors." Oral presentation, competitively selected, Purdue SURF Research Symposium, 2025.
- Islam, M., Kim, J.Y., Yoo, G.H., Im, S., Calderon Ortiz, G., **Ashfaq, H.**, Park, E.S., Hwang, J. "In-Depth Analysis of Structural Heterogeneity in High Entropy Bulk Metallic Glasses Using 4D-STEM." Poster, competitively selected, Microscopy & Microanalysis 2024 Conference, Cleveland, Ohio.

PROFESSIONAL EXPERIENCE

American Electric Power Aug 2025 – Dec 2025

Data Analytics Project Consultant *Columbus, OH*

- Developed predictive models in Python and R forecasting peak electricity demand across PJM's 13-state grid using real-time load data and NOAA weather integration, optimizing capacity-cost management for retail customers.
- Analyzed forecast accuracy across 7-day lead times to set customer alert thresholds, balancing prediction reliability against actionable notification timing, and built Tableau and PowerBI dashboards monitoring peak-hour probabilities across utility zones.

Invicta Counseling May 2023 – Aug 2023

Automation Engineering Intern (Remote) *Islamabad, Pakistan*

- Built an automated data-analysis tool in RStudio and SQL that cut manual processing 40%, automated inventory management through Python and API integration reducing errors 70%, and ran C and C++ system diagnostics that lowered downtime 45%.

AWARDS & HONORS

Best Senior Capstone Project, Department of Data Analytics · Departmental Honor Cords, Computer Science and Data Analytics · Denison Phi Society Inductee, top 7% of Class of 2026 · Denison Alumni Award · Denison Department Scholarship, Music (Percussion) · Purdue SURF Research Fellowship · Intel-funded OSU SURE-5 Research Fellowship.

RELEVANT COURSEWORK

Machine Learning & Computing: Machine Learning; Deep Learning Theory; Algorithm Design & Analysis; Data Structures; Mathematical Foundations of CS; Computer Organization; Software Engineering.

Statistics & Optimization: Applied Statistics & Probability; Time Series & Analysis; Advanced Predictive Methods; Advanced Prescriptive Methods; Linear Programming; Mathematical Modeling; Linear Algebra & Differential Equations; Calculus II, III.

LANGUAGES

Urdu (native) · English (fluent) · Hindi (fluent) · Punjabi (fluent) · Arabic and Persian (read and write) · French (elementary).